

Table 1: How AI enhances edge computing services

Feature	Description	AI-based approach
Low latency	Minimising the delay in data transmission for real-time applications	Implementing AI algorithms for edge processing, reducing round-trip times
Network slicing	Dividing a physical network into multiple virtual networks for different use cases	AI-driven dynamic network slicing for optimised resource allocation based on demand
Massive IoT connectivity	Connecting a vast number of IoT devices efficiently	AI-powered IoT device management for intelligent connectivity and resource allocation
Enhanced mobile broadband	Providing higher data rates for improved mobile connectivity	AI for optimising spectrum usage, predictive maintenance, and bandwidth allocation
Beamforming	Focusing radio signals in specific directions to enhance signal strength	AI algorithms for dynamic beamforming adjustments based on user locations and network conditions
Network security	Protecting against cyber threats and ensuring data integrity	AI-based anomaly detection, predictive analysis, and real-time threat identification
Resource allocation	Efficiently distributing network resources based on demand	Machine learning for dynamic resource allocation, optimising performance in real-time
Energy efficiency	Minimising energy consumption for sustainable network operation	AI-driven energy management, optimising power usage based on traffic patterns and demand
Quality of service (QoS)	Ensuring reliable and high-quality communication services	AI-based monitoring and adaptive adjustments for maintaining optimal QoS levels
Network management	Overseeing and optimising the overall network infrastructure	AI-driven autonomous network management for self-healing, configuration, and optimisation
Predictive maintenance	Anticipating and preventing network equipment failures	AI algorithms analysing historical data for predictive maintenance, reducing downtime
Edge computing integration	Incorporating edge computing for faster data processing at the network edge	AI for efficient task offloading, distributed processing, and edge-to-cloud collaboration
Dynamic network optimisation	Adapting the network in real-time to changing conditions and demands	AI-driven algorithms continuously optimising network parameters based on evolving scenarios

noise might prevent it from learning your true preferences.

However, there are a few things that can be done to protect your data in the federated learning arena.

- **Secure aggregation:** Imagine devices contributing updates to a model without revealing individual details. Secure aggregation protocols like

homomorphic encryption ensure only the combined update is visible, safeguarding individual contributions.

- **Differential privacy:** Imagine adding controlled noise to your emails before contributing them to a spam filter model. This ensures even if the model is compromised, it's impossible to link any specific

email back to you.

- **Federated transfer learning:** Instead of starting from scratch, pre-trained models on public datasets can be used as a base, reducing reliance on sensitive user data. This is like learning a new language by building upon your existing knowledge of a similar one.